

## STUDENT CASE STUDY EXECUTION REPORT

|                |                          |
|----------------|--------------------------|
| Department     | FED                      |
| Subject        | DSA-2                    |
| Academic Year  | I-III Semester (2025-26) |
| Faculty Name   | Razia mam                |
| Student Name   | Garikipati Keerthi       |
| Roll Number    | 2520030416               |
| Branch         | CSE                      |
| Course Outcome | CO-4                     |

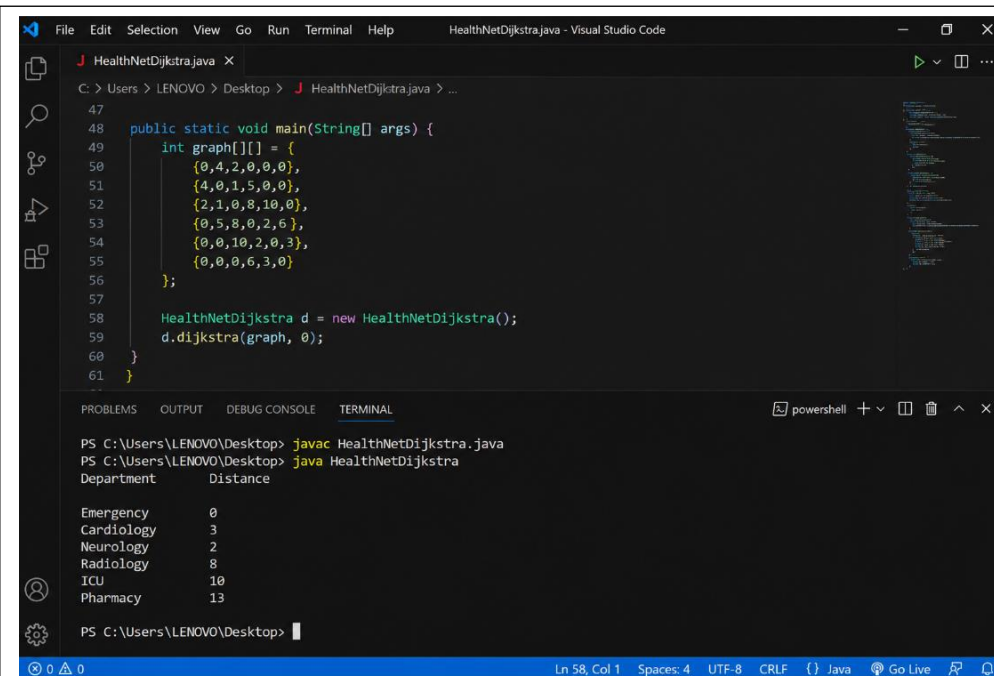
Case Study Topic: Healthnet – Intelligent Hospital Management System

### Work Done Summary:

In this case study, we implemented Dijkstra's Algorithm to find the shortest path between hospital departments in the HealthNet Intelligent Hospital Management System. Each department was represented as a node, and the distance between departments was represented as weighted edges. The algorithm efficiently calculated the shortest route from the Emergency Department to all other departments. This helps optimize patient transfers, emergency services, and resource allocation within the hospital. We compared Dijkstra's Algorithm with traditional traversal methods and observed improved efficiency in finding shortest paths.

### Implementation Details:

1. Represented hospital departments as graph vertices.
2. Connected departments using weighted edges representing travel distances.
3. Implemented Dijkstra's Algorithm using an adjacency matrix.
4. Calculated shortest distances from the Emergency Department.
5. Selected the minimum-distance unvisited vertex at each step.
6. Updated shortest paths to neighboring departments.
7. Displayed the shortest distance from the source department to all other departments.
8. Achieved  $O(V^2)$  time complexity using adjacency matrix implementation.



```

47
48 public static void main(String[] args) {
49     int graph[][] = {
50         {0,4,2,0,0,0},
51         {4,0,1,5,0,0},
52         {2,1,0,8,10,0},
53         {0,5,8,0,2,6},
54         {0,0,10,2,0,3},
55         {0,0,0,6,3,0}
56     };
57
58     HealthNetDijkstra d = new HealthNetDijkstra();
59     d.dijkstra(graph, 0);
60 }
61 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```

PS C:\Users\LENOVO\Desktop> javac HealthNetDijkstra.java
PS C:\Users\LENOVO\Desktop> java HealthNetDijkstra
Department Distance
Emergency 0
Cardiology 3
Neurology 2
Radiology 8
ICU 10
Pharmacy 13
PS C:\Users\LENOVO\Desktop>

```

**Github link:**

**Student signature**

**Faculty signature**